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BCA II Sem. Main/Back Exam 2025-26

BCA–203—Oops Concepts with C++

Time : 3 Hours] **[Max. Marks : 70**

Instructions to Candidates :

Part–A : Short Answer Type Questions (up to 25 words) 10 × 2 = 20 marks. All 10 questions are compulsory.

Part–B : Analytical/Problem Solving questions 5 × 4 = 20 marks. Candidates have to answer 5 questions out of 7.

Part–C : Descriptive/Analytical/Problem Solving questions 3 × 10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form no. 205).

1 _____ **2** _____

Part–A

1. Write any four differences between C and C++.
2. Explain compilation process of a C++ program.
3. List any four basic data types in C++.
4. What is a reference variable ?
5. Consider following code snippet. What will be value of variable c after executing this code. Assurte all headers are included :

```
int a = 5, b= 3, c;
c = a++ + --b * 2 / (a - 4) + b;
```

6. How does break and continue statements differ ?
7. Give any two differences between structure and union with example.
8. What is the use of destructor ?
9. What do you understand by function overloading ?
10. What is an abstract class ?

Part–B

1. Explain various constraints on constructors and destructors.
2. Write a C++ program to demonstrate the concept of an array of objects and accessing their data members using member functions.
3. What are manipulators in C++ ? Explain with examples.
4. What is runtime polymorphism ? Explain with the help of virtual functions.
5. Write a program to demonstrate the use of inline functions in C++.
6. What is a copy constructor ? How is it different from a default constructor ?
7. Explain any two types of inheritance with suitable examples.

Part–C

Write a C++ program that takes a student's marks (out of 100) as input :

(i) Use an if-else statement to check whether the student has passed (pass marks : 40).

(iii) If missing data is to be assumed to develop the graph based on the following range:

(a) 20 – 100: Count 34

(b) 101 – 180: Count 52

(c) 181 – 260: Count 65

(d) 261 – 340: Count 70

(iv) If fabsL (int a)vs. f(int a)

Write a C++ program to perform the following: (05 Questions)

(a) Create a file named 'Student.txt'

(b) Accept and store student data (roll number, total marks, and name) into the file using ofstream.

(c) Read and display the contents of the file using ifstream

(d) Discuss the role of index in the server-fetched stored and linked functions. How do they reduce fragmentation

Why must lowercase i be permitted in overloaded << operator for a class Complex that stores complex numbers

Your program should

(a) Have a constructor to initialize complex numbers

(b) Overload the + operator using a member function

(c) Use a member function to read values, accept the results

(vii) Compare new (dynamic) and delete (dynamic) arrays arrays this operator overloading

(viii) What is ? compare its demonstrate parameter passing techniques in function

(ix) Call by Value

(x) Call by Reference using reference variable

(xi) Suppose an int c/cube function to be saving for integers using each technique, and show the differences

Your report must reflect these results.

Also, briefly explain how the behavior differs in both methods.

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